Southern New Hampshire University

CS 405 Project Two Script Template

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1	Hello, my name is Jose Alonso. This presentation explains the security policies developed for Green Pace. The goal is to protect the company's software, data, and systems by using secure coding practices, encryption, access controls, testing, and automation.
2	**Defense in Depth** Defense in depth means using multiple layers of security instead of relying on just one method. Green Pace can protect its systems by combining secure coding, encryption, authentication, authorization, testing, and monitoring. If one security measure fails, the other layers can help reduce the damage.
3	**Threats Matrix** The threats matrix identifies possible security problems, their likelihood, and their potential impact. Examples include invalid input, SQL injection, memory errors, unauthorized access, and problems with array boundaries. These threats can lead to data exposure, system crashes, or unauthorized changes. Green Pace should focus first on threats that could cause serious damage and are more likely to happen. **Threats that are Likely to happen are** Data Value Validation, String Correctness and Bounds, Validate External Input **The Priority threats are** Prevent SQL Injection and Memory Protection and Lifetime **The Low-Priority threats are** Use Assertions for Internal Invariants and Exception and Error Handling

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	The Unlikely threats are Data Type Safety, initialize Variables Before Use, and Array and Container Bounds
4	# Ten Secure Coding Principles The ten secure coding principles help developers write safer and more reliable software. Green Pace's standards cover important practices such as using safe data types, validating information, checking string lengths, preventing SQL injection, and managing memory. Developers should also handle errors correctly, initialize variables, and check array boundaries. Following these principles helps prevent common coding mistakes and reduces security risks. The ten secure coding principle are • Data Type Safety • Data Value Validation String Correctness and Bounds • Prevent SQL Injection • Memory Protection and Lifetime • Use Assertions for Internal Invariants • Exception and Error Handling • Validate External Input • Initialize Variables Before Use • Array and Container Bounds
5	# Coding Standards Green Pace's coding standards explain how developers should apply secure coding practices when writing software. Developers must validate data, prevent unsafe database queries, manage memory carefully, and handle errors correctly. They should also initialize variables and check array boundaries to prevent unexpected behavior. These standards provide developers with consistent rules

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	to follow during development and code reviews. Using these rules can reduce coding mistakes and help prevent security vulnerabilities.
6	## Encryption Policies Encryption protects sensitive information from unauthorized access. Green Pace should encrypt sensitive data stored in databases, files, and backups. Data sent across networks should use secure protocols such as TLS. Access to encryption keys should be restricted, and keys should be protected from exposure. The company should also limit access to sensitive information while it is being used and avoid storing passwords or secrets in application logs.
7	## Triple-A Policies Triple-A stands for authentication, authorization, and accounting. Authentication verifies a user's identity. Authorization controls which files, systems, and actions a user can access. Accounting records important activities so the company can review events and investigate security incidents. Green Pace should use unique accounts, multifactor authentication for administrators, least-privilege access, and protected audit logs. These controls help prevent unauthorized access and improve accountability.
8	## Unit Testing For this slide, I used Microsoft Visual Studio Test Explorer to demonstrate unit testing for the Green Pace secure coding standards. The tests focus on checking

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	that a C++ vector collection works correctly. They check whether values can be added, whether the collection can be resized or cleared, and whether the collection reports its size and capacity correctly. The tests also check how the program handles invalid indexes. For example, the `AtThrowsOutOfRange` and `InvalidIndexThrowsException` tests verify that trying to access an index outside the vector's valid range throws an exception. This helps prevent errors when a program tries to access data that does not exist. Unit testing is important because it helps developers find problems early, before the software is released. These tests support the secure coding standard [STD-010-CPP], which focuses on array and container bounds. The results in Visual Studio Test Explorer provide evidence of which tests pass or fail and help developers identify areas that need correction.
9	# Automation Summary Automation allows Green Pace to run security checks throughout software development. Automated unit tests can check program behavior, while static analysis tools can identify possible coding problems. Dependency scanning can identify vulnerable libraries, and secret scanning can help detect exposed credentials. These checks should run during development and before release. When a problem is found, developers should fix it and run the checks again.
10	# Tools Green Pace can use several tools to improve software security. Visual Studio supports code development and unit testing. Cppcheck and Clang-Tidy can help identify potential coding problems. AddressSanitizer can detect certain memory errors. Dependency scanning tools help identify vulnerable libraries, while secret

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	scanning tools look for exposed credentials. Using these tools together provides additional ways to find and fix problems before software reaches users.
11	## Risks and Benefits Using secure coding standards has many benefits for Green Pace. These standards help prevent security problems, protect sensitive information, and make the software more reliable. They can also help developers find mistakes earlier, which can save time and money. However, there are some challenges. Developers may need extra time to learn the standards, test their code, and use security tools. Fixing security problems can also take time and delay development. Even with these challenges, the benefits are worth it because security problems can be more expensive to fix after software is released. Overall, following secure coding standards helps Green Pace reduce risks, improve software quality, and protect its users. By making security part of the development process from the beginning, the company can build safer and more dependable software.
12	## Recommendations I recommend that Green Pace require security checks before software is released. Developers should follow all ten coding standards, perform code reviews, and run automated tests. The company should also scan dependencies and source code for vulnerabilities and exposed secrets. User permissions should be reviewed regularly, encryption keys should be protected, and security issues should be tracked until they are fixed. Green Pace should review its policies at least once a year.

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13	# Conclusions In conclusion, secure coding standards are important for protecting Green Pace's software and its users. By following these standards, developers can reduce security risks, prevent common coding mistakes, and make the software more reliable. Using tools like Visual Studio and Cppcheck, along with unit testing, can help developers find problems before the software is released. Encryption and proper access controls also help protect sensitive information from unauthorized access. Overall, security should be part of every step of software development, not something added at the end. By following these recommendations, Green Pace can build safer software, protect user data, and reduce the chances of future security problems.
14	This final slide lists the sources used to support the security policies and recommendations in this presentation. These sources provide information about secure coding practices, software security, encryption, and testing tools. They help explain why following security standards is important for Green Pace. Using reliable sources helps make sure that the recommendations are based on trusted security guidance. These references can also help developers learn more about protecting sensitive information, preventing common coding errors, and building more secure software.

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	These sources support the goal of improving Green Pace's software security and helping developers follow secure coding practices throughout the development process. Thank you for listening to my presentation. These policies provide Green Pace with a practical starting point for improving software security and protecting its systems and data.